SEEDLINE · PROJECT PLAN

SeedLine project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 pipeline, kit 1.9.1 upgrade): workstreams, resources, stage gates, next steps and the reference list

SeedLine project plan

SeedLine is at stage P3 (design outputs, TRL 3) with the kit 1.9.1 pipeline documents and a detailed model of both concept options. The next gate, P4 (verification, TRL 4), is on hold under the portfolio's TRL cap of 3. The decision requested from Amish is the concept choice (SDL-DEC-001: option A, the current design, recommended) and approval of the fixes made under his standing instruction of 2026-10-23 (F1 to F8), which he can reverse. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and requirements	Design lead	PRB, STM, OPT, URS, PRS, RTM	Done at P3 (2026-10-24); concept choice awaiting Amish
Calculations	Design lead	SDL-CAL-001 (`docs/04-calcs/sizing.py`), STEM check script	Done at P3; R1, R2 and R10 at risk; push endurance to assess
CAD and drawings	Design lead	`cad/src/model.py` at D3 for options A and B, STEP, SDL-DWG-001 and making sketches SDL-DWG-101 to 124	Done at P3 (`cad/fidelity.json`)
Build plan and decisions	Design lead	SDL-BLD-001, SDL-DEC-001	Done at P3
Regulatory, risk and market	Design lead	SDL-REG-001, SDL-RSK-001, SDL-MKT-001	First issue; items "to assess" listed
Co-design partner	Project lead (Amish)	Partner agreement, co-design sessions, soil data	Not started; first candidate named (SDL-DEC-001)
Bench and field tests	Test lead	TST documents: metering bench test, push test, fatigue test	On hold (P4, TRL cap 3)
Renders and storefront	Coordinator	`media/options.png` , photoreal renders, cards	Scenes exported; rendered on Amish's Mac

Resources needed

Table 2. Resources (costs are the BOM estimates where a BOM line exists; others to assess).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts (P4)	One prototype with both kits	Purchasing and printing	266.10 (BOM estimate)
Workshop time (P4)	Frame, opener, handle and marker made to SDL-BLD-001	About 1.5 h of cutting and drilling (in BOM line 12) plus assembly, about 1 to 2 days	In BOM line 12; assembly to assess
Desktop FDM printer time (P4)	Plates, hopper, housing, guard, liners, knob	About 1.5 to 2 h per plate; hopper, housing and guard to assess	Printer time not costed
Bench test rig (P4)	Sticky-belt metering test to ISO 7256-1 terms	To assess	To assess
Partner field trial (P5)	Field counts, push and depth on partner soils	To assess with the partner	To assess
Regulatory review	Items "to assess" in SDL-REG-001	To assess	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (kit 1.8 documents added 2026-10-24)
P1 Definition	URS, PRS, RSK, MKT	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check and feasibility closed	Met, except the STEM check: open (rows S6 to S10, S16 and S17 need bench, print, weighing or ergonomic data; option B rows OB4 and OB6 open) and the concept choice awaiting Amish (SDL-DDR-004)
P3 Design outputs	Model at D3, STEP, DWG, priced BOM, BLD, DEC, design FMEA, traceability complete	Met (2026-10-24)
P4 Verification	Bench tests of metering, push and fatigue; verification report; RSK updated	On hold (TRL cap 3)
P5 Validation	Partner field trials, human factors, certification status in REG	On hold
P6 Commercialization	Launch plan with a named deployment partner	On hold

Next steps and decision requested

1. Amish: decide the concept (SDL-DEC-001 D1; option A recommended, 169 against 131 of 230 weighted points) and review the eight fixes made under the standing instruction (F1 to F8).
2. Coordinator: render `media/options.png` and the product renders from the exported scenes on Amish's Mac.

3. When the TRL cap lifts: metering bench test (PR-001, PR-002), push and endurance assessment (PR-010, ISO 11228-2), plate wear and handle fatigue tests.
4. Approach the first candidate partner and replace the assumed soils, crops and spacings with partner data.

References

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